

Building a model of a (growing/shrinking) epidemic

Parasites are awesome. That is a fact, but also a motivation for our activity today. Let's think about what a parasite "wants"¹ to do. Taking a parasite's eye view², the "goal" is to leave a lot of offspring that in turn leave a lot of offspring. However, it is less important to replicate within or on one's host—these copies would be competitors for resources and anyway are a sort of dead end—than it is to cause new infections in new hosts. Indeed, we have seen and briefly discussed this idea in the guise of R_0 , which we can define as the number of secondary infections caused by a single infected individual over the span of that infection³. Today, our goal is to think of the processes that lead to new infections (gains) and those that cause infections to end (losses).

Putting together a model

Models, whether mathematical or box and arrow, can be terrifically useful ways to organize our thinking about a process or system. They force us to consider the players, the steps involved, the forces at work, and what we're willing to ignore and what seems essential. So we are going to work together to create a model of a spreading infection.

Wait! Don't freak out! We're only going to think in boxes and arrows⁴ (and logic). And I will be here to help you out. Let's give it a try.

¹ Major scare quotes here. No one, to my knowledge, has figured out how to talk to a parasite to ask them what they want. Really, we are making assumptions based on first principles.

² Like a bird's eye view, but (usually) one from inside a host.

³ ...in a wholly susceptible population. But let's leave that off the table for the moment.

⁴ Importantly, these box-and-arrow diagrams are *not* DAGs! Rather, they represent gains and losses.

It is usually helpful to think of a specific infection in a specific population. Be sure you and your group are on the same page about this specific system before proceeding.

Gaining infections

I would like you to take this space to write out the steps or processes involved in an infected host spreading that infection to susceptible hosts⁵ in the population. Who are the players? What are the processes? Is there more than one pathway?

⁵ A fancy name for individual animals (or plants if you're into that thing) that are not yet infected, but are infectable.

Have a good list? Let's now draw this process in a box-and-arrow format on the whiteboard. (Or if there is a format that makes more sense to you, use that!)

Losing infections

Now write out the steps or processes involved in the population *losing* infected hosts. Who are the players? What are the processes? Is there more than one pathway?

Try to add these boxes or arrows to the diagram you have already drawn. Do you need to add new boxes?

Adding terms

Odds are reasonably good that you now have a complex diagram on the white board with a lot of boxes or a lot of arrows. Congratulations! You know (a little bit about) a lot of biology! However, models are much easier, and often more helpful, when they are simple. So let's see if we can simplify things a bit.

Please demarcate (circle, underline, box-in, or whatever is helpful) and label these different processes on your diagram.

- Transmission
- Recovery
- Virulence (=disease-induced death)

These are the basic processes involved in most models of infectious disease spread. But they may not cover all of the steps you included in your diagram. So step back and think about what else there should be in a good model of this spreading infection.

Let's regroup and discuss

We will spend a bit of time as a class comparing notes. I will also guide you all towards a particular sort of formulation. However, I want you to know that your models are also perfectly reasonable and potentially quite useful! There is no "right" model of infectious disease spread (or most anything else). Our goal was to organize our thinking, and hopefully yours did. I am going to push you towards my⁶ version for two reasons: 1) my version is very useful and has a lot of lessons to teach, and 2) we need to have a common language for talking about epidemics moving forward. In short, my version will be more useful for a lot of reasons that have nothing to do with the "correctness" or realism of any of these models. Thanks for playing along!

⁶ OK, not mine. I'm really borrowing from a long line of people that developed these ideas over the last century.